Report

Haoyu Huang

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0.1 Step 1

0.1.1 Drawing the Box Plots

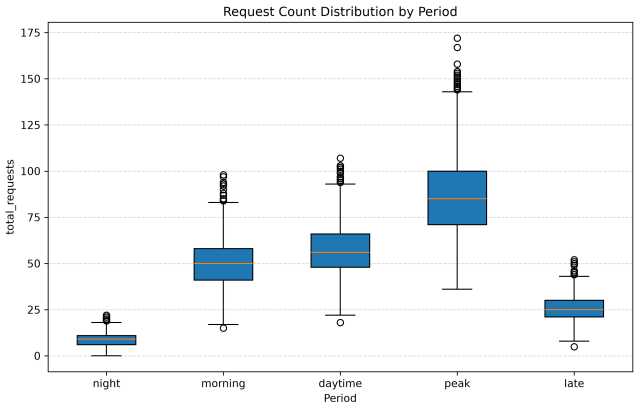


Figure 1: Box plots of text requests

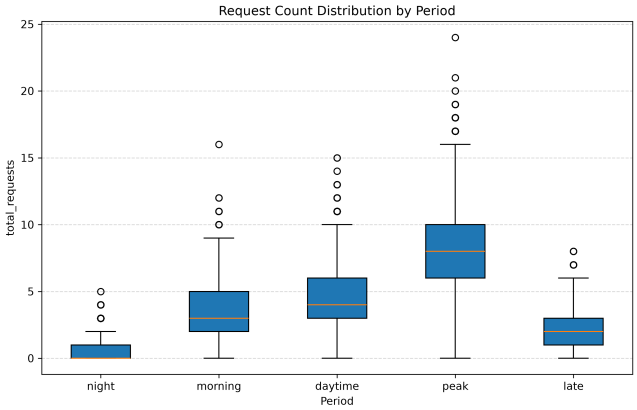


Figure 2: Box plots of image requests

The box plots for the two types of requests are presented in fig. 1 and fig. 2, respectively. It is evident that an analysis restricted to peak hours on weekdays alone cannot adequately illustrate the actual trends in request volumes.

0.1.2 Establish Poisson distribution

Poisson distributions are assumed for N_t and N_i, specifically,

$$P(N_t = k) = \frac{\lambda_t^k e^{-\lambda_t}}{k!},$$

$$P(N_i = k) = \frac{\lambda_i^k e^{-\lambda_i}}{k!}.$$

The parameters λ_t and λ_i can be estimated using the sample mean:

$$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} N_i.$$

The 95% confidence interval for the Poisson parameter λ can be approximated using the normal approximation:

$$CI_{95\%} = \left[\hat{\lambda} - 1.96\sqrt{\frac{\hat{\lambda}}{n}}, \hat{\lambda} + 1.96\sqrt{\frac{\hat{\lambda}}{n}}\right].$$

. The estimation result is shown on table 1.

Table 1: Parameters Estimation of Poisson distribution

Type	λ	95% (s)
Text	45.4513	[45.3583, 45.5444]
Image	3.6389	[3.6126, 3.6653]

0.1.3 Validate Poisson Distribution

To validate the adaptability of Poisson distribution, we implemented χ goodness-of-fit test, specifically

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

, where O_i denotes the observed frequency of class i and E_i denotes the expected frequency of class i. The result is presented on table 2

Table 2: Parameters Estimation of Poisson distribution

Type	χ^2	conclusion
Text	8369296.14	Rejected
Image	72418.63	Rejected

0.1.4 Correlation Coefficient of N_t and N_i

The correlation coefficient can be described as

$$r = \frac{\sum_{i=1}^{n}(N_t - \bar{N})(N_i - \bar{N_i})}{\sqrt{\sum_{i=1}^{n}(N_t - \bar{N_t})^2}\sqrt{\sum_{i=1}^{n}(N_i - \bar{N_i})^2}}$$

. The result is approximately equal to 0.7851, which suggest a strong relation between text and image requests.

0.2 Step 2

0.2.1 Report for service time

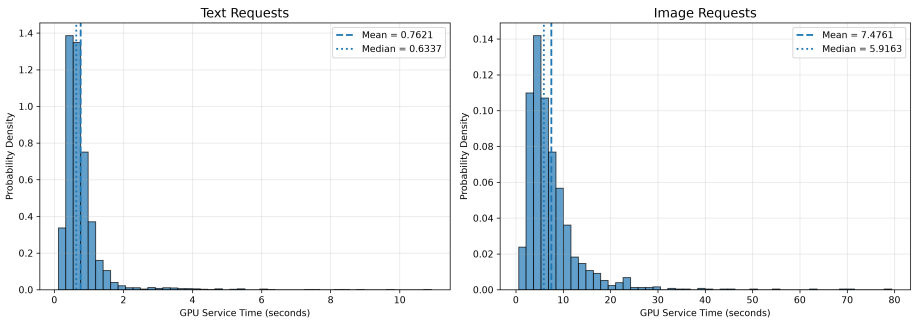


Figure 3: Histogram of two types of requests

Table 3: Descriptive Statistics of Text and Image Request Durations (seconds)

Type	Size	Mean (s)	Std (s)	Median (s)	95% (s)	99% (s)
Text	4000	0.762147	0.628502	0.633650	1.464310	3.528130
Image	1600	7.476124	6.247750	5.916300	17.529665	29.749451

0.2.2 Report for cold start service time

0.2.3 Establish Probability Model

0.2.4 Estimate Model Parameters